Midterm Practice Questions · Answer Key

MATH60033A · Quantitative Methods in International Business

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Open this only after answering the questions on paper. The answers are model answers: on the exam, any precise formulation earns the marks, and **working is marked** — a correct derivation with an arithmetic slip earns most of them.

Part A — Definitions and short answers

A1. *(easy)* The mean is the sum divided by n, the balance point; the median is the middle value once sorted. The median is robust: an extreme value moves the mean (it enters the sum) but not the middle position.

A2. *(easy)* The mean computed after dropping a fixed share (say 5% or 10%) of observations from each tail. Report it when a few extreme values distort the mean but you want to use more of the data than the median does.

A3. *(medium)* The deviations from $\bar{x}$ sum to zero, because $\bar{x}$ was estimated from the same data: only $n-1$ of them are free. Dividing by n would systematically underestimate the variance.

A4. *(medium)* The SD measures the spread of the **observations**; the SE, $s/\sqrt{n}$, measures the spread of the **mean** across repeated samples. A larger sample shrinks the SE (the mean is estimated more precisely) but not the SD (the observations are as spread out as before).

A5. *(easy)* $IQR = Q_3 - Q_1$, the width of the middle 50% of the data. It ignores both tails, so a long tail does not inflate it the way it inflates the SD.

A6. *(easy)* A long **right** tail: a few large values. They pull the mean above the median.

A7. *(easy)* How heavy the tails are compared with a normal distribution: $g_2 > 0$ means more extreme values than a normal. A normal has excess kurtosis 0.

A8. *(medium)* Not necessarily. The rule holds for a roughly **normal** variable; a symmetric distribution can still have heavy or light tails (check g_2).

A9. *(easy)* The sum of squared residuals, $\sum_i (y_i - \hat{y}_i)^2$. Squaring makes every miss positive, so misses above and below cannot cancel, punishes one large miss more than several small ones, and gives a closed-form solution, $(X^\top X)^{-1} X^\top y$.

A10. *(medium)* "Comparing countries with the same investment, a one-percentage-point higher tertiary education share is associated with a 0.31-percentage-point higher GDP growth." Units, "holding investment fixed", and "associated with", not "causes".

A11. *(difficult)* $\hat{\beta}_j$ equals the slope from regressing the part of y not explained by the other regressors on the part of x_j not explained by them. It is a **partial** relationship: what is left of the association after the other regressors have taken their share.

A12. *(easy)* $R^2 = 1 - SSR/SST$, the share of the variance of y the model accounts for. It does not say the relationship is causal, nor that the model's assumptions hold (a curved cloud can have a high R^2), nor that it will predict new data well.

A13. *(medium)* R^2 never falls when a regressor is added, even a useless one. The adjusted R^2 penalises each extra regressor and falls when the new one improves the fit by less than chance would.

A14. *(easy)* $RSE = \sqrt{SSR/(n-k-1)}$: the typical size of a residual, in the **units of** y — the typical miss of the model.

A15. *(difficult)* Linearity in the parameters, exogeneity $\mathbb{E}[\varepsilon \mid X] = 0$, and no perfect collinearity (full column rank). Constant variance (homoskedasticity) is **not** needed for unbiasedness — only for the classical standard errors and for efficiency.

A16. *(medium)* The variance of the errors changes with the fitted values: the residual plot **fans out**. The coefficients stay unbiased, but the classical standard errors are wrong, so tests and intervals are wrong. Use robust (HC) standard errors.

A17. *(medium)* An **outlier** has a large residual (unusual y given x). A **leverage point** has an unusual x. An **influential point** changes the fit when removed — typically high leverage **and** a large residual.

A18. *(medium)* How much all the fitted values move when that observation is deleted; it combines leverage and residual. Investigate it — find which row it is and why — and report the fit with and without it. Never delete it silently.

A19. *(medium)* $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is from regressing x_j on the other regressors. Above about 10, x_j is almost a combination of the others: its standard error is inflated and its sign can be unstable (multicollinearity).

A20. *(medium)* BIC, because its penalty per parameter, $\ln n$, exceeds AIC's 2 as soon as $n > 7$. The models must be fitted on **the same rows**; otherwise the likelihoods are not comparable.

A21. *(easy)* It can predict probabilities below 0 or above 1, and it forces the effect on the probability to be the same everywhere (a straight line), which is implausible near 0 and 1. Its errors are also heteroskedastic by construction.

A22. *(easy)* odds $= p/(1-p)$. For $p = 0.20$: $0.20/0.80 = 0.25$, "one event for every four non-events".

A23. *(medium)* In **log-odds**: one more unit of x raises the log-odds by 0.26. $e^{0.26} = 1.30$ is the **odds ratio**: the odds are multiplied by 1.30, i.e. 30% higher. The wrong reading: "the **probability** is 30% higher".

A24. *(medium)* The probability is an S-shaped function of x: its slope, $p(1-p)\beta$, depends on where you evaluate it, largest at $p = 0.5$ and small near 0 or 1. A constant odds ratio moves the probability by different amounts depending on the starting point.

A25. *(easy)* The parameter values that make the observed data — here, the 0s and 1s actually observed — as probable as possible.

A26. *(medium)* Wald: $z = \hat{\beta}/SE$, one coefficient at a time, from one model. Likelihood ratio: $LR = 2(\ell_{full} - \ell_{restricted}) \sim \chi^2_q$, comparing two nested models. Use the LR test for several coefficients **jointly**, such as all the dummies of one categorical variable.

A27. *(medium)* $1 - \ell_{model}/\ell_{null}$: how far the model's log-likelihood improves on the intercept-only model. It is not a share of variance, and its values run lower than an OLS R^2; 0.2–0.4 already indicates a good fit.

A28. *(difficult)* The cutpoints $c_1 < c_2 < ...$ are the thresholds between categories, one per cut; they act as intercepts. Proportional odds: each predictor has **one** coefficient, so the same odds ratio applies at every cut. Check it by fitting a binary logit at each cut and comparing the slopes.

A29. *(difficult)* The category every other one is compared with; each equation models $\ln[P(k)/P(\text{baseline})]$. An RRR of 0.80: a one-unit rise in x multiplies the ratio $P(\text{vocational})/P(\text{academic})$ by 0.80 — 20% fewer vocational students per academic one. It does not say the probability of vocational falls by 20%.

A30. *(difficult)* Discrimination: does the model **rank** positives above negatives (AUC)? Calibration: when the model says 20%, does the event happen about 20% of the time? AUC depends only on the ranking, so a model can rank perfectly while every probability is, say, twice too high — useless for pricing or thresholds.

A31. *(easy)* With a 2% base rate, predicting "never" is 98% accurate and finds nothing. Precision $= TP/(TP+FP)$: of those flagged, the share that are positive. Recall $= TP/(TP+FN)$: of the positives, the share flagged.

A32. *(medium)* Test error $= \text{bias}^2 + \text{variance} + \text{noise}$. With few rows and many correlated predictors, OLS is unbiased but its variance is enormous; a little bias that cuts the variance a lot lowers the total.

A33. *(medium)* It makes one in-or-out decision at a time, each depending on the previous ones; resampling the rows changes the path and the final model, and noise variables get in. The selection reflects this sample, not a property of the world, and the p-values computed afterwards are invalid.

A34. *(medium)* It shrinks every coefficient toward zero, more where predictors are collinear, but never to exactly zero. $X^\top X + \lambda I$ is invertible for any $\lambda > 0$, even when $X^\top X$ is singular ($p > n$).

A35. *(medium)* It sets weak coefficients to **exactly zero** (the corners of its diamond-shaped budget), so it selects variables. Limits: with $p > n$ it keeps at most n variables; among correlated predictors it keeps one, almost arbitrarily; the selected set is not "the true model".

A36. *(easy)* A ridge component: correlated predictors are kept or dropped **together** instead of one standing in for the group — the **grouping effect**.

A37. *(difficult)* The penalty compares coefficient sizes, which depend on units: unstandardised, a variable in euros and one in thousands of euros are punished differently. Inside each fold, because standardising on all rows lets the held-out fold's information into the training, which makes the CV score optimistic (leakage).

A38. *(medium)* λ_{min} minimises the CV error. λ_{1se} is the largest λ whose error is within one standard error of that minimum. It gives a simpler (more shrunk) model whose error is indistinguishable from the best.

A39. *(difficult)* FE remove everything constant within a country over time (geography, history, institutions); the cost is that the effect of any time-invariant variable cannot be estimated, and only within-country variation identifies the slopes. RE assume the country effects are uncorrelated with the regressors. The Hausman test compares the two;

choose on whether the RE assumption holds, never on which gives the significant answer.

A40. *(medium)* Observations of the same country are correlated over time, so the sample carries less independent information than n suggests. Cluster the standard errors by country.

Part B — Calculations

B1. *(easy)* **(a)** Mean $= 35/5 = 7$; sorted $4, 6, 7, 8, 10$, median $= 7$. **(b)** Deviations $-3, 1, -1, 3, 0$; squares $9, 1, 1, 9, 0$, sum 20; $s^2 = 20/4 = 5$, $s = 2.236$. **(c)** $SE = 2.236/\sqrt{5} = 1.00$.

B2. *(easy)* **(a)** $SE = 12/\sqrt{36} = 2$; interval $50 \pm 1.96 \times 2 = (46.08,\ 53.92)$. **(b)** $n = 144$: $SE = 12/12 = 1$, halved; s stays about 12 — more data does not make the observations less spread out.

B3. *(medium)* **(a)** $g_1 = \frac{1}{36} \cdot \frac{1{,}166{,}400}{30^3} = \frac{1{,}166{,}400}{972{,}000} = 1.20$; threshold $2\sqrt{6/36} = 0.816 \rightarrow$ substantially **right**-skewed, consistent with mean > median. **(b)** $g_2 = \frac{1}{36} \cdot \frac{116{,}640{,}000}{30^4} - 3 = 4.00 - 3 = 1.00$; threshold $4\sqrt{6/36} = 1.633 \rightarrow$ not substantially kurtic. **(c)** The median (105): the right tail pulls the mean up.

B4. *(medium)* **(a)** $\bar{x} = 3$, $\bar{y} = 4$; $\sum(x-\bar{x})(y-\bar{y}) = 4+0+0+0+2 = 6$, $\sum(x-\bar{x})^2 = 10$; $\hat{\beta}_1 = 0.6$, $\hat{\beta}_0 = 4 - 0.6 \times 3 = 2.2$. **(b)** Fitted $2.8, 3.4, 4.0, 4.6, 5.2$; residuals $-0.8, 0.6, 1.0, -0.6, -0.2$ (they sum to 0); $SSR = 0.64+0.36+1+0.36+0.04 = 2.4$. **(c)** $SST = 4+0+1+0+1 = 6$; $R^2 = 1 - 2.4/6 = 0.60$; $RSE = \sqrt{2.4/3} = 0.894$.

B5. *(easy)* **(a)** $t = 0.85/0.34 = 2.50$; interval $0.85 \pm 0.666 = (0.184,\ 1.516)$; significant at 5% ($|t| > 1.96$; the interval excludes 0). **(b)** $\bar{R}^2 = 1 - 0.60 \times 49/45 = 0.347$.

B6. *(medium)* **(a)** Bias $= \beta_z \delta_1 = -0.3 \times 0.5 = -0.15$: negative, the short regression **understates** β_1. **(b)** $0.45 = \beta_1 - 0.15$, so $\beta_1 = 0.60$.

B7. *(medium)* **(a)** $\sum h_{ii} = p = 3$; average $3/40 = 0.075$. **(b)** Yes: 0.21 is 2.8 times the average, above the usual $2p/n = 0.15$ flag. High leverage alone is not influence: check its residual and its Cook's distance. **(c)** $4/n = 0.10$.

B8. *(easy)* $VIF = 1/(1 - 0.92) = 12.5$: the coefficient's variance is 12.5 times what it would be with uncorrelated regressors, its SE $\sqrt{12.5} \approx 3.5$ times larger. Above 10: a problem.

B9. *(medium)* **(a)** $AIC_A = 420 + 8 = 428$, $AIC_B = 410 + 14 = 424$; $BIC_A = 420 + 4 \times 4.605 = 438.4$, $BIC_B = 410 + 7 \times 4.605 = 442.2$. **(b)** AIC prefers B (lower), BIC prefers A. BIC charges $\ln 100 = 4.6$ per parameter instead of 2, so B's three extra parameters cost more than its gain in fit.

B10. *(medium)* **(a)** $x^\top \hat{\beta} = -3.0 + 0.04 \times 50 + 0.9 = -0.1$; odds $= e^{-0.1} = 0.905$; $p = 0.905/1.905 = 0.475$. **(b)** $e^{0.9} = 2.46$: a foreign-owned firm has 2.46 times the odds of exporting of an otherwise identical domestic firm (146% higher odds). **(c)** $p(1-p)\beta = 0.475 \times 0.525 \times 0.04 = 0.0100$: about **1.0 percentage point** per point of productivity, at this firm.

B11. *(medium)* **(a)** Odds $0.20/0.80 = 0.25$; new odds $0.25 \times 1.5 = 0.375$; new probability $0.375/1.375 = 0.273$. **(b)** +7.3 percentage points (from 20% to 27.3%), a 36% relative rise. An odds ratio multiplies the **odds**, not the probability.

B12. *(medium)* **(a)** $LR = 2(-412.6 + 418.9) = 12.6 > 7.81$ (χ^2_3): the three dummies jointly improve the model; reject the null that all three are zero. **(b)** $1 - 412.6/450 = 0.083$.

B13. *(easy)* **(a)** $\tau^\star = c_{FP}/(c_{FP} + c_{FN}) = 1/(1+9) = 0.10$. **(b)** $0.15 > 0.10$: flag. With 0.5 the applicant would not be flagged — the default threshold ignores the costs.

B14. *(difficult)* $P(Y \le 1) = F(1.0 - 1.5) = F(-0.5) = 0.378$; $P(Y \le 2) = F(3.0 - 1.5) = F(1.5) = 0.818$. So $P(1) = 0.378$, $P(2) = 0.818 - 0.378 = 0.440$, $P(3) = 1 - 0.818 = 0.182$ (they sum to 1).

B15. *(medium)* Denominator $e^0 + e^{-0.5} + e^{0.2} = 1 + 0.607 + 1.221 = 2.828$. $P(\text{academic}) = 1/2.828 = 0.354$, $P(\text{general}) = 0.607/2.828 = 0.215$, $P(\text{vocation}) = 1.221/2.828 = 0.432$.

B16. *(easy)* **(a)** $TP = 30$, $FP = 70$, $FN = 20$, $TN = 880$. **(b)** Accuracy $= 910/1000 = 0.91$; precision $= 30/100 = 0.30$; recall $= 30/50 = 0.60$. **(c)** $950/1000 = 0.95$ — higher than the model's accuracy, although it finds no failure at all.

B17. *(difficult)* **(a)** $\det = 24 - 9 = 15$; $(X^\top X)^{-1} = \frac{1}{15}\begin{pmatrix} 4 & -3 \\ -3 & 6 \end{pmatrix}$; $\hat{\beta}^{OLS} = \frac{1}{15}(48 - 30,\ -36 + 60) = (1.200,\ 1.600)$. **(b)** $X^\top X + 2I = \begin{pmatrix} 8 & 3 \\ 3 & 6 \end{pmatrix}$, $\det = 39$; $\hat{\beta}^{ridge} = \frac{1}{39}(72 - 30,\ -36 + 80) = (1.077,\ 1.128)$. **(c)** $1 - 1.077/1.2 = 10.3\%$ and $1 - 1.128/1.6 = 29.5\%$.

B18. *(medium)* **(a)** Subtract 0.25 from $|\rho_j|$, keep the sign, set to 0 if negative: $(0.35,\ -0.10,\ 0,\ 0,\ 0.03)$ — variables 1, 2 and 5 selected. **(b)** Threshold $\lambda\alpha = 0.125$, divisor $1 + 0.25 \times 0.5 = 1.125$: $(0.475,\ -0.225,\ 0.025,\ 0,\ 0.155)/1.125 = (0.422,\ -0.200,\ 0.022,\ 0,\ 0.138)$ — variables 1, 2, 3 and 5. **(c)** The elastic net keeps one more (variable 3): its lower

threshold lets weaker predictors in, and its ridge part shrinks the survivors proportionally.

B19. *(medium)* $\lambda_{min} = 0.1$ (error 1.90). One-SE bound $1.90 + 0.11 = 2.01$. The largest λ with error ≤ 2.01 is 0.5 (1.98; at 1.0 the error 2.05 is above). $\lambda_{1se} = 0.5$: more shrinkage, so a smaller model, with an error indistinguishable from the best.

B20. *(easy)* **(a)** Effect $= 3.0 - 2 \times 0.05\,x = 3.0 - 0.1\,x$: +2.0 at 10 years, −1.0 at 40. **(b)** $x^\star = 3.0/0.1 = 30$ years (a maximum, since the square term is negative). **(c)** 3.0 is the effect only at $x = 0$; with a square term the effect depends on x.

B21. *(medium)* **(a)** Men: 1.5 per year; women: $1.5 - 0.5 = 1.0$ per year. **(b)** Gap (women − men) $= 4 - 0.5\exp$: +4 at 0 years, $4 - 5 = -1$ at 10. **(c)** $4 - 0.5\exp = 0$ at 8 years.

B22. *(easy)* Classical: $t = 3.75$, interval (0.143, 0.457). Clustered: $t = 1.67$, interval (−0.053, 0.653). Report the clustered one: the panel's observations are not independent. The estimate did not change; what you may claim did — no longer significant at 5%.

B23. *(easy)* $14.2 > 7.81$ (χ^2_3): reject. FE and RE differ by more than sampling noise, so the RE assumption fails — use fixed effects.

B24. *(easy)* Four dummies; the omitted level is the reference. With five plus an intercept, the dummies sum to the intercept column: perfect collinearity (the dummy-variable trap), and the model cannot be estimated.

Part C — Diagnosis

C1. *(medium)* The empirical rule is applied to a variable that is clearly skewed: income cannot be negative, and an SD of 71% of the mean signals a long right tail. Report the median and the IQR instead, and check g_1 against its threshold.

C2. *(easy)* "Not significant" is read as "no effect". With $p = 0.31$ the data cannot distinguish the effect from zero, but the interval almost certainly also includes large negative values. Report the interval and say the evidence is inconclusive.

C3. *(medium)* Unbiased coefficients do not save the inference: under heteroskedasticity the classical standard errors, t-statistics and intervals are wrong. Use robust (HC) standard errors.

C4. *(medium)* AIC compared on different rows: 1,200 against 900. Likelihoods on different samples are not comparable, and the drop mostly reflects fewer observations. Refit both on the same 900 rows.

C5. *(medium)* An odds ratio read as a probability ratio (relative risk), and the conclusion is that collateral raises default risk. 1.40 means the **odds** are 40% higher; the change in probability depends on the baseline. (And a positive association with collateral may reflect which borrowers are asked to provide it — not causation.)

C6. *(medium)* Accuracy on a 1% outcome: "never fraud" is 99% accurate. Report the base rate, precision and recall. And the threshold ignores the costs: $\tau^\star = 1/(1 + 50) \approx 0.02$, not 0.5.

C7. *(difficult)* Three errors. Standardising on **all** rows before cross-validation (leakage). Treating the lasso's selection as the true model. Reporting OLS p-values on the selected columns: the same data chose and then tested them, so the p-values are invalid. Fix: put the scaler inside a pipeline, report the selected set and its out-of-sample error, no post-selection p-values.

C8. *(difficult)* The specification is chosen by which gives the significant result — specification search. Choose on the assumption: run the Hausman test (or argue whether the country effects can be correlated with the regressor). If RE is rejected, FE is the estimate, insignificant or not.

Part D — Judgement (what a strong answer contains)

D1. *(medium)* **What it is:** a cross-country association, €410 per index point, holding nothing else fixed that we know of. **Not licensed:** "will raise": richer countries can afford digital government (reverse causality), and institutions or education can drive both (omitted variables); $R^2 = 0.58$ says nothing about cause. **Evidence needed:** a design that isolates the change — within-country change over time with fixed effects, a staggered rollout compared with non-adopters, controls for the obvious confounders. **Meanwhile:** report the association with its interval, and do not attach a euro figure to the policy.

D2. *(difficult)* **What it is:** a predictive association in this bank's data. **Not licensed:** region may stand in for income, sector or collateral; refusing a whole region also raises fairness and legal issues. **Evidence:** does the odds ratio survive controls for the borrower's own risk factors? Is the model calibrated for that region? **Meanwhile:** price risk on the borrower's characteristics, use a cost-based threshold, and review the variable's legitimacy before using it.

D3. *(medium)* **What it is:** six variables that jointly predict well at one λ, on this sample. **Not licensed:** a dropped variable is not "no effect" — the lasso keeps one of a correlated group almost arbitrarily, and with $p > n$ it can keep at most n. **Evidence:** selection stability across resamples, an elastic net to see whether groups enter together, a causal design for any variable whose effect matters. **Meanwhile:** report the predictive performance, not a list of "drivers".

D4. *(difficult)* Fixed effects use only **within-firm** variation: the estimate answers "when **this** firm raises its training budget, does **its** productivity rise?" It removes everything constant about each firm, including whatever makes some firms both train more and be more productive. The question "are high-training firms more productive than others?" is a **between-firm** comparison that FE deliberately throws away; answering it needs the cross-sectional variation, with its own confounding problems.

D5. *(easy)* The mean is pulled up by a few very large firms: $g_1 = 6.4$ is far above any threshold, and the median (18) is a fourteenth of the mean. The typical firm has about 18 employees — small, not medium. Report the median and the IQR, say the distribution is strongly right-skewed, and model the log of employment if it enters a regression.

Questions: `PRACTICE.md` · *the exam:* `EXAM.md` *and its solutions* · *the summary:* `MIDTERM-SUMMARY.pdf`